

THE GEORGE WASHINGTON UNIVERSITY
Department of Statistics
STAT 4197/6197 Syllabus
Spring, 2026

Course and Contact Information

Course: STAT 4197-80/6197-80, Fundamentals of SAS Programming for Data Management
Meeting Time: Friday, 12:45-3:15 pm
Class Meeting Location: Rome Hall, Room B104

Instructor

Name: Pradip K. Muhuri
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Office hours: Fridays 3:30-4:30 pm

Course Description

The course will introduce students to SAS programming for accessing, managing, manipulating, and summarizing data. Specifically, it will cover how to write SAS programs to control, modify, aggregate, combine, and reshape SAS data sets. Additionally, the course will cover the SAS macro facility, as well as the main aspects of the Structured Query Language (SQL) and the SAS/IML (Interactive Matrix Language). Students will also learn about selected SASPy module applications in Python sessions, exploring the two-way data exchange between SAS data sets and Python Pandas Data Frames. Finally, the course will briefly compare data-wrangling techniques in SAS and Python.

Course Prerequisites

STAT 1051 or STAT 1053 and prior programming knowledge.
Students who have already received credit for STAT 4197 are not eligible to receive credit for STAT 6197.

Learning Outcomes

Upon successful completion of this course, students will be able to:

- read raw data and Microsoft Excel files into SAS data sets,
- manipulate data using SAS expressions, functions, arrays, and Do Loops,
- aggregate, combine, reshape, and summarize data using DATA and PROC steps,
- automate and customize the generation of SAS code using the macro facility,
- manipulate matrices and SAS data sets using PROC IML and
- run analytics with a Python interface to the SAS System in Jupyter notebooks.

Textbooks and Online SAS Documentation

Delwiche L, and Slaughter S. The Little SAS Book: A Primer. 2019. Sixth Edition. Cary, NC: SAS Institute Inc.

Ottesen RA, Delwiche LD, and Slaughter SJ. *Exercises and Projects for The Little SAS Book*. 2020. Sixth Edition Paperback. Cary, NC: SAS Institute Inc.

Cody, R. *Cody's Data Cleaning Techniques Using SAS®*, Third Edition - March 2017

SAS/IML 14.3 User's Guide (https://documentation.sas.com/doc/en/pgmsascdc/9.4_3.3/imlug/titlepage.htm)

SASPy (<https://sassoftware.github.io/saspy/>)

Schedules of Lectures and Assessments

Week	Date	Lecture Topic	Assessment
1	01/16/2026	The SAS System, Interfaces, and Basic Concepts	
2	01/23/2026	DATA Step: Reading Data and Creating Reports	
3	01/30/2026	Working with Formats/Informats and Transforming Data	
4	02/06/2026	Functions, Data Conversions, Do Loops, and Arrays	Test 1
5	02/13/2026	Controlling and Managing SAS Data Sets	Homework Assignment 1 Given
6	02/20/2026	Aggregating Data and Combining SAS Data Sets (DATA Step vs. PROC SQL)	
7	02/27/2026	Exploring and Summarizing Data and Generating Reports (SAS Procedures and Output Delivery System)	Test 2
8	03/06/2026		Midterm Exam
	03/09/2026 – 03/14/2026	Spring Break (No class)	
9	03/20/2026	SAS Macro Language Basics	Homework Assignment 1 Due Homework Assignment 2 Given (for STAT 6197 students only)
10	03/27/2026	Macro Functions and Working with Macros	
11	04/03/2026	Additional Topics on Macro Facility	Test 3
12	04/10/2026	Matrix Operations and Functions in SAS/IML	Test 4 Homework Assignment 2 Due (for STAT 6197 students only)
13	04/17/2026	SAS Programming Efficiency and Miscellaneous	Test 5 (Optional)
14	04/24/2026	Applications of the SASPy Module Using Jupyter Lab	
15	05/08/2026 (Tentative date)		Final Exam

Average Amount of Learning Expected Per Week

Students will receive 2.5 hours of classroom instruction and are expected to spend at least 5 hours on independent learning per week, totaling a minimum of 7.5 hours per week.

STAT 4197 Grading Policy

Students enrolled in STAT 4197 will complete seven required assessments, excluding the optional fifth test. The percentage weight for each assessment type is shown in parentheses below.

- Four tests (quizzes or programming exercises) (20%)

- One homework assignment (10%)
- Midterm exam (35%)
- Final exam (35%).

STAT 6197 Grading Policy

Students enrolled in STAT 6197 will complete eight required assessments, excluding the optional fifth test. The percentage weight for each assessment type is shown in parentheses below.

- Four tests (quizzes or programming exercises) (20%)
- Two homework assignments (20%)
- Midterm exam (30%)
- Final exam (30%).

If a student enrolled in STAT 4197 or STAT 6197 elects to take the optional fifth test, the lowest score among the five tests will be dropped for grading purposes.

The overall weighted average points will be calculated using the weights assigned to the assessment categories mentioned above. The final letter grades for the course will be determined based on the numerical range of the overall weighted average points, as set by the instructor. An incomplete grade may be given only to students who have passed the course but cannot complete it due to well-documented circumstances beyond their control.

Class Policies

- Students should attend all classes and engage fully in class activities.
- No late submission of assignments/tests is acceptable (unless for serious reasons with documentation).
- There will be no make-up exams/tests or extra-credit assignments.
- Students must turn off their mobile devices during class sessions and exams.

Software and Other Tools

SAS® software options:

- Option 1: Use SAS in the windowing environment on your Windows laptop (if you have one) by requesting a SAS software license from the GW Instructional Technology Lab (ITL). To do this, fill out an online form (<https://it.gwu.edu/sas>) and follow the installation instructions provided by ITL. Note that the licensed SAS software is only compatible with Windows laptops, not MacBooks. If this option is not feasible for you, consider exploring options 2, 3, or 4.
- Option 2: Access SAS in the windowing environment on GW classroom computers. No software installation is required.
- Option 3: Use a free version of SAS known as SAS OnDemand for Academics (ODA) in a cloud-based environment. To do this, you must first create a SAS profile (https://www.sas.com/en_us/software/on-demand-for-academics.html) and then register for SAS ODA. No software installation is necessary.
- Option 4: Use SAS in the windowing environment on the GW virtual computer lab (<https://gwu.apporto.com>). No software installation is needed.

Blackboard: Refer to the lecture notes and assignment descriptions that have been posted on Blackboard.

GitHub Repository: Accept the instructor's invitation to join the GitHub organization. Once you have done this, you will be able to access the sample SAS programs used in each lecture session from the instructor's private GitHub repository.

University Policy on Observance of Religious Holidays

Under University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. See "Religious Holidays" at <https://provost.gwu.edu/policies-procedures-and-guideline> for details.

Academic Integrity Code

Academic integrity is essential to the educational process, and GW takes these matters very seriously. Violations of academic integrity include cheating in any form, such as misrepresenting one's work, taking credit for the work of others without giving proper credit, and fabricating information. Additional information can be found at the Office of Academic Integrity website: <https://studentconduct.gwu.edu/academic-integrity>. For further assistance, don't hesitate to get in touch with rights@gwu.edu or call 202-994-6757.

Generative Artificial Intelligence (GAI) Policies

Students can utilize generative artificial intelligence (GAI) tools, such as ChatGPT, to enhance their comprehension of the SAS code covered in class. It is important to note that the use of GAI tools is prohibited during in-class tests and exams. However, students are allowed to use a double-sided, one-page "Cheat Sheet" for these assessments.

Safety and Security

In an emergency, call GWPD at 202-994-6111 or 911. For situation-specific actions, review the Emergency Response Handbook at <https://safety.gwu.edu/emergency-response-handbook>. Get Out, Hide Out, or Take Out in an active violent situation. Stay informed: <https://safety.gwu.edu/stay-informed>.

Disability Support Services (DSS)

Any student in need of an accommodation based on the potential impact of a disability should contact the Disability Support Services office at 202-994-8250 in the Marvin Center, Suite 242. This is to establish eligibility and coordinate reasonable accommodations.

Mental Health Services 202-994-5300

The University Counseling Center (UCC) offers 24/7 assistance and referral to address students' personal, social, career, and study skills problems. Student services include crisis and emergency mental health consultations, confidential assessments, counseling services (individual and small group), and referrals.

University policies, academic support, student support, and campus emergency information

Information on University policies, academic support, support for students in and outside of the classroom, and GW campus emergency information can be found at the following link: bulletin.gwu.edu/university-syllabus-policies/.